

DIGITAL SAT MATH MASTERY SERIES

CREATING LINEAR MODELS FROM WORD PROBLEMS

Total Questions: 50 | Easy: 8 | Medium: 20 | Hard: 22

DOCUMENT 1 — PRACTICE QUESTIONS

Aligned to Digital SAT Question Style & Cognitive Demand

All questions original — fresh numbers, fresh contexts

Topic	Creating Linear Models from Word Problems
Document	DOCUMENT 1 — PRACTICE QUESTIONS
Questions	50
Difficulty	Easy: 8 Medium: 20 Hard: 22
Format	Multiple Choice (4 options)
Exam Target	Digital SAT (DSAT)

■■■ CREATING LINEAR MODELS FROM WORD PROBLEMS — PRACTICE QUESTIONS ■■■

Instructions: Choose the best answer for each question. All questions are original and written in authentic Digital SAT style. Difficulty level is noted for each question.

Q1. *DIFFICULTY: EASY*

A taxi company charges a flat fee of \$3.50 plus \$1.75 per mile driven. Which of the following equations represents the total cost C , in dollars, for a ride of m miles?

- A) $C = 1.75m$
- B) $C = 3.50m + 1.75$
- C) $C = 1.75m + 3.50$
- D) $C = 3.50 + m$

Q2. *DIFFICULTY: EASY*

A cell phone plan costs \$25 per month plus \$0.10 per text message sent. If Priya sends t text messages in a month, which equation gives her total monthly bill B , in dollars?

- A) $B = 0.10t$
- B) $B = 25t + 0.10$
- C) $B = 0.10t + 25$
- D) $B = 25 + t$

Q3. *DIFFICULTY: EASY*

Marcus is saving money for a video game. He already has \$18 saved and plans to save \$6 each week. Which of the following equations models the total amount A , in dollars, Marcus has saved after w weeks?

- A) $A = 6w$
- B) $A = 18w + 6$
- C) $A = 6w + 18$
- D) $A = 18 + w$

Q4. *DIFFICULTY: EASY*

A candle is 12 inches tall when lit. It burns at a rate of 0.5 inches per hour. Which equation gives the height h , in inches, of the candle after t hours of burning?

- A) $h = 12t - 0.5$
 - B) $h = 0.5t + 12$
 - C) $h = 12 - 0.5t$
 - D) $h = 0.5 - 12t$
-

Q5. DIFFICULTY: **EASY**

A car rental company charges \$40 per day plus a one-time insurance fee of \$15. Which equation represents the total cost C , in dollars, for renting a car for d days?

- A) $C = 40d$
 - B) $C = 40 + 15d$
 - C) $C = 40d + 15$
 - D) $C = 55d$
-

Q6. DIFFICULTY: **EASY**

A swimming pool holds 8,000 gallons of water. Water drains from the pool at a rate of 200 gallons per hour. Which equation represents the volume V , in gallons, remaining after h hours of draining?

- A) $V = 200h$
 - B) $V = 8000h - 200$
 - C) $V = 8000 + 200h$
 - D) $V = 8000 - 200h$
-

Q7. DIFFICULTY: **EASY**

Kenji earns \$12.50 per hour working at a bakery. Last week he also received a \$30 tip. Which equation gives his total weekly earnings E , in dollars, if he worked h hours last week?

- A) $E = 12.50h$
 - B) $E = 30h + 12.50$
 - C) $E = 12.50h - 30$
 - D) $E = 12.50h + 30$
-

Q8. DIFFICULTY: **EASY**

A store sells notebooks for \$2.75 each. A customer also pays a one-time membership fee of \$10. Which equation gives the total amount T , in dollars, a member pays when buying n notebooks?

- A) $T = 2.75n$
 - B) $T = 10n + 2.75$
 - C) $T = 2.75n + 10$
 - D) $T = 12.75n$
-

Q9. DIFFICULTY: **MEDIUM**

A water tank initially contains 500 liters. Water flows in at 30 liters per minute and flows out at 12 liters per minute. Which equation represents the volume V , in liters, in the tank after t minutes?

- A) $V = 500 + 18t$
 - B) $V = 500 - 18t$
 - C) $V = 500 + 42t$
 - D) $V = 500 - 42t$
-

Q10. DIFFICULTY: **MEDIUM**

A plumber charges \$75 for the first hour and \$55 for each additional hour. Which equation gives the total charge C , in dollars, for a job lasting h hours, where $h > 1$?

- A) $C = 75h + 55$
 - B) $C = 55h + 75$
 - C) $C = 55h + 20$
 - D) $C = 130h$
-

Q11. DIFFICULTY: **MEDIUM**

The cost of a school fundraiser dinner is \$200 for venue rental plus \$8.50 per ticket sold. The school collected \$624 in ticket revenue. Which equation could be used to find the number of tickets t sold?

- A) $200 + 8.50t = 624$
 - B) $8.50t = 624$
 - C) $200t + 8.50 = 624$
 - D) $8.50t - 200 = 624$
-

Q12. DIFFICULTY: **MEDIUM**

A rocket sled starts at rest and accelerates uniformly. After 3 seconds it is traveling at 90 feet per second, and after 7 seconds it is traveling at 210 feet per second. Which equation best models the speed s , in feet per second, as a linear function of time t , in seconds?

- A) $s = 30t$
 - B) $s = 30t + 90$
 - C) $s = 30t - 90$
 - D) $s = 90t + 30$
-

Q13. DIFFICULTY: **MEDIUM**

A temperature reading in Celsius C and the corresponding Fahrenheit reading F satisfy a linear relationship. When $C = 0$, $F = 32$, and when $C = 100$, $F = 212$. Which equation converts C to F ?

- A) $F = 1.8C + 32$
 - B) $F = 1.8C - 32$
 - C) $F = 32C + 1.8$
 - D) $F = 0.8C + 32$
-

Q14. DIFFICULTY: **MEDIUM**

Two friends each start reading a book on the same day. Anika starts on page 1 and reads 22 pages per day. Ben starts on page 50 (he had read ahead) and reads 15 pages per day. Which equation gives the total pages p each has read after d days as a system in terms of Anika's pages?

- A) $p = 22d + 1$
 - B) $p = 22d$
 - C) $p = 15d + 50$
 - D) $p = 22d - 1$
-

Q15. DIFFICULTY: **MEDIUM**

A company rents office space. The monthly rent r , in dollars, is modeled by $r = 1,200 + 4.50f$, where f is the number of square feet rented. What is the best interpretation of 4.50 in this context?

- A) The base monthly rent for any office
 - B) The total monthly rent for 1 square foot
 - C) The additional monthly cost per square foot of office space
 - D) The maximum possible monthly rent
-

Q16. DIFFICULTY: **MEDIUM**

An athlete's heart rate H , in beats per minute, during a workout session is modeled by $H = 72 + 5.4m$, where m is the number of minutes into the workout. What does the value 72 most likely represent?

- A) The athlete's heart rate at the start of the workout
 - B) The increase in heart rate each minute
 - C) The maximum heart rate achievable
 - D) The time when heart rate equals zero
-

Q17. DIFFICULTY: **MEDIUM**

A catering service charges by the number of guests. For a party with 80 guests, the charge is \$960. For a party with 120 guests, the charge is \$1,360. Assuming a linear relationship, which equation models the charge C in terms of the number of guests g ?

- A) $C = 10g + 160$
 - B) $C = 8g + 160$
 - C) $C = 12g + 160$
 - D) $C = 10g + 200$
-

Q18. DIFFICULTY: **MEDIUM**

A diver descends from the surface at a constant rate. After 2 minutes the diver is at a depth of 18 meters, and after 5 minutes the diver is at 45 meters. Which equation models depth d , in meters, as a function of time t , in minutes?

- A) $d = 9t$
 - B) $d = 9t - 18$
 - C) $d = 18t$
 - D) $d = 9t + 2$
-

Q19. DIFFICULTY: **MEDIUM**

The equation $P = 4.20n - 380$ models the monthly profit P , in dollars, for a bakery that sells n loaves of bread per month. According to this model, how many loaves must the bakery sell in order to break even ($P = 0$)?

- A) About 90 loaves
 - B) About 95 loaves
 - C) About 76 loaves
 - D) About 80 loaves
-

Q20. DIFFICULTY: **MEDIUM**

A landscaping crew earns a set fee of \$120 plus \$35 per hour for additional labor. If a second crew earns only \$45 per hour with no set fee, and both crews work j hours, which equation represents the condition under which the first crew charges less than the second?

- A) $120 + 35j < 45j$
 - B) $120 + 35j > 45j$
 - C) $35j < 120 + 45j$
 - D) $120 - 35j < 45j$
-

Q21. DIFFICULTY: **MEDIUM**

The function $f(x) = -3.2x + 96$ models the amount of fuel f (in gallons) remaining in a truck's tank after x hours of driving. What is the most reasonable interpretation of the x -intercept of this function?

- A) The initial amount of fuel in the tank
 - B) The rate at which fuel is consumed
 - C) The number of hours until the tank is empty
 - D) The fuel remaining after one hour
-

Q22. DIFFICULTY: **MEDIUM**

A gym membership costs \$50 to join and \$30 per month. A competing gym has no joining fee but costs \$40 per month. After how many months m will the total cost of both gyms be equal?

- A) $m = 3$
 - B) $m = 4$
 - C) $m = 5$
 - D) $m = 6$
-

Q23. DIFFICULTY: **MEDIUM**

A new apartment building has 200 units. The property manager expects that d units will be vacant each month due to tenant turnover. After 6 months, only 164 units are occupied. Which equation best models the number of occupied units O after t months?

- A) $O = 200 - 6t$
 - B) $O = 200 - 36t$
 - C) $O = 164 + 6t$
 - D) $O = 200 + 6t$
-

Q24. DIFFICULTY: **HARD**

A factory produces widgets at a cost of \$3.80 per widget plus fixed daily costs of \$420. The factory sells each widget for \$6.50. Which of the following equations gives the factory's daily profit P , in dollars, when w widgets are produced and sold?

- A) $P = 6.50w - 420$
 - B) $P = 2.70w - 420$
 - C) $P = 3.80w + 420$
 - D) $P = 2.70w + 420$
-

Q25. DIFFICULTY: **HARD**

The linear model $V = 24,000 - 1,800t$ represents the value V , in dollars, of a car t years after purchase. A second car is purchased at the same time for \$18,000 and depreciates at \$1,200 per year. How many years after purchase will the two cars have the same value?

- A) $t = 4$
 - B) $t = 5$
 - C) $t = 6$
 - D) $t = 10$
-

Q26. DIFFICULTY: **HARD**

A scientist measures the mass of a radioactive sample every hour. The data suggest a linear relationship for the first 10 hours: at $t = 1$ the mass is 47.6 grams, and at $t = 4$ the mass is 40.4 grams. Which linear equation best models mass m in terms of time t ?

- A) $m = -2.4t + 50$
 - B) $m = -2.4t + 40$
 - C) $m = 2.4t + 50$
 - D) $m = -7.2t + 50$
-

Q27. DIFFICULTY: **HARD**

Company A charges \$0.15 per unit plus a monthly fee of \$180. Company B charges \$0.25 per unit with no monthly fee. For what number of units u is Company A's monthly cost less than Company B's?

- A) $u < 1,200$
 - B) $u < 1,800$
 - C) $u > 1,200$
 - D) $u > 1,800$
-

Q28. DIFFICULTY: **HARD**

The equation $C = 0.08s + 1,200$ models the monthly operating cost C , in dollars, of running a delivery service, where s is the number of packages delivered that month. A rival service charges a flat \$1,680 per month regardless of volume. How many packages per month makes the rival service cheaper?

- A) More than 5,000
 - B) More than 6,000
 - C) More than 7,000
 - D) Less than 6,000
-

Q29. DIFFICULTY: **HARD**

An employee receives a base salary of \$3,200 per month plus a commission of 4% on all sales above \$10,000. Which of the following equations gives the employee's monthly earnings E , in dollars, when monthly sales total s dollars and $s > 10,000$?

- A) $E = 3,200 + 0.04s$
 - B) $E = 3,200 + 0.04(s - 10,000)$
 - C) $E = 3,200 + 0.04(s + 10,000)$
 - D) $E = 0.04s - 3,200$
-

Q30. DIFFICULTY: **HARD**

A train travels at a constant speed. At 9:00 AM it is 310 miles from its destination, and at 11:30 AM it is 185 miles from its destination. Which linear equation gives the distance d , in miles, from the destination as a function of time t in hours after 9:00 AM?

- A) $d = 310 - 50t$
 - B) $d = 310 + 50t$
 - C) $d = 185 - 50t$
 - D) $d = 50t - 310$
-

Q31. DIFFICULTY: **HARD**

The monthly profit P of a small restaurant is modeled by $P = 18.50n - 3,700$, where n is the number of meals served. If the restaurant is open 30 days per month, approximately how many meals per day must it serve to achieve a monthly profit of at least \$1,850?

- A) At least 15 meals per day
 - B) At least 10 meals per day
 - C) At least 22 meals per day
 - D) At least 42 meals per day
-

Q32. DIFFICULTY: **HARD**

A linear function passes through (2, 11) and (6, 3). What is the y-intercept of this function?

- A) 15
 - B) 13
 - C) 17
 - D) 9
-

Q33. DIFFICULTY: **HARD**

On a highway, speed limit signs are posted every k miles. The number of signs N within a 240-mile stretch is modeled by $N = 240/k$. If the equation were rewritten as a linear model for total sign-reading time T (seconds), where each sign takes 2 seconds, and $k = 3$, which expression gives T ?

- A) $T = 160$
 - B) $T = 480$
 - C) $T = 80$
 - D) $T = 240$
-

Q34. DIFFICULTY: **HARD**

A manufacturer produces two products. Product A yields \$5 profit per unit and Product B yields \$8 profit per unit. The total weekly profit is \$2,480. If the manufacturer produces 160 units of Product A, how many units of Product B are produced?

- A) 200
 - B) 210
 - C) 180
 - D) 220
-

Q35. DIFFICULTY: **HARD**

Water evaporates from a reservoir at a linear rate. On Day 0, the reservoir contains 4,500 acre-feet. On Day 12, it contains 3,780 acre-feet. Which equation gives the volume V , in acre-feet, after d days?

- A) $V = 4,500 - 60d$
 - B) $V = 4,500 + 60d$
 - C) $V = 3,780 - 60d$
 - D) $V = 4,500 - 720d$
-

Q36. DIFFICULTY: **MEDIUM**

A local theater sells adult tickets for \$14 and student tickets for \$8. At one performance, 220 tickets were sold for a total revenue of \$2,480. Which system of equations could be used to find the number of adult tickets a and student tickets s sold?

- A) $a + s = 220$ and $14a + 8s = 2,480$
 - B) $a + s = 2,480$ and $14a + 8s = 220$
 - C) $a - s = 220$ and $14a + 8s = 2,480$
 - D) $a + s = 220$ and $8a + 14s = 2,480$
-

Q37. DIFFICULTY: **MEDIUM**

Natalie walks from home to school at 4 miles per hour and her brother drives the same route at 30 miles per hour. Natalie leaves 20 minutes before her brother. Which equation gives Natalie's distance d from home after t hours?

- A) $d = 4t$
 - B) $d = 4(t + 1/3)$
 - C) $d = 30t$
 - D) $d = 4t - 20$
-

Q38. DIFFICULTY: **MEDIUM**

A fitness app tracks a user's daily step count. On Monday the user took 7,200 steps, and each subsequent day the user increased steps by 300. Which linear equation gives the step count S on day n , where $n = 1$ represents Monday?

- A) $S = 300n + 7,200$
 - B) $S = 300n + 6,900$
 - C) $S = 7,200n + 300$
 - D) $S = 300n - 7,200$
-

Q39. DIFFICULTY: **MEDIUM**

A moving company charges \$90 per hour for a crew of 3 workers plus \$0.50 per pound for items above 2,000 pounds. A move takes 4 hours and the items weigh 2,800 pounds. What is the total charge?

- A) \$760
 - B) \$760 plus tax
 - C) \$400
 - D) \$1,000
-

Q40. DIFFICULTY: **MEDIUM**

The cost C , in dollars, to print custom t-shirts at a shop is $C = 8.25n + 45$, where n is the number of shirts. Which statement best describes the 45 in this equation?

- A) The cost per additional shirt
 - B) The total cost for 45 shirts
 - C) A fixed setup fee charged regardless of how many shirts are printed
 - D) The maximum number of shirts that can be ordered
-

Q41. DIFFICULTY: **HARD**

Two workers are painting a house. Worker A has already painted 120 square feet and paints at 40 square feet per hour. Worker B starts at the same time as Worker A begins counting but paints at 55 square feet per hour from scratch. After how many hours will Worker B have painted more square feet than Worker A?

- A) After 8 hours
 - B) After 9 hours
 - C) After 7 hours
 - D) After 10 hours
-

Q42. DIFFICULTY: **HARD**

The linear model $W = 185 - 2.5t$ represents a person's weight W in pounds t weeks after starting a diet. What is the most reasonable interpretation of the y-intercept?

- A) The person's weight after 2.5 weeks
 - B) The rate of weight loss per week
 - C) The person's weight at the start of the diet
 - D) The total number of weeks on the diet
-

Q43. DIFFICULTY: **HARD**

An elevator starts at floor 28 and descends at a constant rate of 2 floors every 3 seconds. A second elevator starts at floor 4 and ascends at a constant rate of 1 floor every 2 seconds. Which equation gives the floor number F of the descending elevator after s seconds?

- A) $F = 28 - (2/3)s$
 - B) $F = 28 + (2/3)s$
 - C) $F = 28 - 2s$
 - D) $F = (2/3)s - 28$
-

Q44. DIFFICULTY: **HARD**

A school district's budget B , in millions of dollars, follows the model $B = 42 + 1.8y$, where y is the number of years after 2015. According to this model, in which year will the budget first exceed \$60 million?

- A) 2024
 - B) 2025
 - C) 2026
 - D) 2027
-

Q45. DIFFICULTY: **HARD**

A lab starts an experiment with 5 grams of a substance. The substance dissolves at a constant rate. After 8 minutes, 1.8 grams remain. Which linear equation models the amount A , in grams, remaining after m minutes?

- A) $A = 5 - 0.4m$
 - B) $A = 5 + 0.4m$
 - C) $A = 1.8 + 0.4m$
 - D) $A = 5 - 0.8m$
-

Q46. DIFFICULTY: **HARD**

A delivery service offers two pricing plans. Plan X: \$2.50 per pound with no base charge. Plan Y: \$1.80 per pound plus a \$14 base charge. For what weight w , in pounds, is Plan Y cheaper than Plan X?

- A) $w > 20$
 - B) $w < 20$
 - C) $w > 14$
 - D) $w > 10$
-

Q47. DIFFICULTY: **HARD**

The function $f(t) = 3,600 - 45t$ models the distance in meters between a hiker and a mountain peak t minutes after the hiker begins the descent. What does the value 3,600 represent?

- A) The speed of the hiker in meters per minute
 - B) The time it takes to reach the base
 - C) The distance from the peak at the start of the descent
 - D) The distance remaining after one minute
-

Q48. DIFFICULTY: **HARD**

A phone manufacturer's revenue R , in millions, is modeled by $R = 12n - 180$, where n is the number of phones (in thousands) produced and sold. What is the minimum number of phones, in thousands, that must be sold for the revenue to be positive?

- A) 14
 - B) 15
 - C) 16
 - D) 18
-

Q49. DIFFICULTY: **HARD**

An investor puts \$5,000 in Account A earning \$200 per year (simple interest) and \$3,000 in Account B earning \$150 per year. After y years, which equation gives the combined value V of both accounts?

- A) $V = 8,000 + 350y$
 - B) $V = 5,000 + 150y + 3,000 + 200y$
 - C) $V = 350y$
 - D) $V = 8,350y$
-

Q50. DIFFICULTY: **HARD**

A car accelerates from 20 mph to 65 mph over 9 seconds. Assuming constant acceleration, which linear equation models the speed S , in mph, after t seconds?

- A) $S = 5t + 20$
 - B) $S = 45t + 20$
 - C) $S = 5t + 65$
 - D) $S = 20 + 9t$
-